

2024

COMPUTER SCIENCE AND ENGINEERING

Paper : CSEL-1012

(Natural Language Processing)

Full Marks : 70

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer **Question no. 1**, **Question no. 2**, and **any four** from the rest.

1. Answer **any five** questions :

2×5

- (a) What is word tokenization? Describe some issues related to tokenization.
- (b) Explain why CFG is used to represent natural language in parsing.
- (c) Identify whether the different meanings arise from structural, semantic, or pragmatic ambiguity for each sentence :
 - (i) Time flies like an arrow.
 - (ii) He crushed the key to my heart.
- (d) Define RNN. State three architectures of RNN.
- (e) Differentiate between open-class words and closed-class words with examples.
- (f) Differentiate between top-down and bottom-up parsing.
- (g) What is the importance of syntax in NLP?

2. Answer **any five** questions :

4×5

- (a) Draw the syntax tree (top-down) parsing for the following sentences :
 - (i) The restaurant serves hot dosa.
 - (ii) The dog is crying loudly.
- (b) Perform the POS tag sequence for the following phrase, 'the old man,'
Given that we have the following probabilities :
 $P(\text{the}|\text{Det}) = 0.5,$
 $P(\text{the}|\text{Noun}) = 0.1,$
 $P(\text{old}|\text{Noun}) = 0.002,$
 $P(\text{old}|\text{Adj}) = 0.001,$
 $P(\text{old}|\text{Verb}) = 0.05,$

Please Turn Over

$P(\text{man}|\text{Noun}) = 0.6,$
 $P(\text{man}|\text{Verb}) = 0.01$
 $P(\text{man}|\text{Det}) = 0.00001,$
 $P(\text{Noun}|\text{Det}) = 0.3,$
 $P(\text{Adj}|\text{Det}) = 0.2,$
 $P(\text{Noun}|\text{Noun}) = 0.2,$
 $P(\text{Adj}|\text{Noun}) = 0.006,$
 $P(\text{Noun}|\text{Adj}) = 0.3,$
 $P(\text{Noun}|\text{Verb}) = 0.4,$
 $P(\text{Verb}|\text{Noun}) = 0.2,$
 $P(\text{Verb}|\text{Adj}) = 0.004,$
 $P(\text{Verb}|\text{Verb}) = 0.01.$

Work out the steps of the Viterbi algorithm in detail. You can use a Tableto show the steps. Assume all other conditional probabilities not mentioned to be zero. Also, assume that all tags have the same probability of appearing at the beginning of a sentence.

(c) Explain the process of machine translation using encoder-decoder architecture.

(d) Solve the following problem using Bayes theorem :

A study of pleas and prison sentences found that 45% of the subjects studied were sent to prison. Among those sent to prison, 40% chose to plead guilty. Among those not sent to prison, 55% chose to plead guilty.

- (i) If one of the study subjects is randomly selected, find the probability of getting someone who was not sent to prison.
- (ii) If a study subject is randomly selected and it is then found that the subject entered a guilty plea, find the probability that this person was not sent to prison.

(e) Identify the following sentence segments' head and morphological type (Noun Phrase, Verb Phrase, Adjective Phrase, Adverbial Phrase) :

- (i) The president of the company
- (ii) Looked up the chimney
- (iii) Angry as a hippo
- (iv) Rapidly like a bat
- (v) Man with a golden arm.

(f) Explain the working principle of LSTM (Long-Short-Term-Memory) with a focus on RNNs. Draw the diagram and the equations.

(g) Define Perplexity. Explain the Shannon-Visualization Model with respect to Perplexity.

3. Derive a top-down, depth-first, left-to-right parse tree for the given sentence :

The angry bear chased the frightened little squirrel.

S → NP VP	Det → the
NP → Det Nom	Adj → little angry frightened
VP → V NP	N → squirrel bear
Nom → Adj Nom N	V → chased

10

4. Derive the shift-reduce parser for the following sentence after finding the syntactic relationship between each word.

The man beat the boy with a hard stick.

10

5. 50% of people don't smoke, 20% are light smokers and 30% are heavy smokers. Heavy smokers are twice as likely to die prematurely as light smokers and light smokers are twice as likely to die prematurely as non-smokers. What is the probability of being a heavy smoker, given that a person died prematurely?

10

6. You are given a multilingual corpus and tasked with building a dependency parser that performs well across multiple languages. Discuss the strategies you would employ to handle linguistic diversity, considering both supervised and unsupervised learning techniques.

10

7. (a) Define dependency graph.

- (b) Write down the formal conditions for a dependency graph.

- (c) Given the following CFG grammar, perform syntactic analysis of the following sentence using any parsing method.

2+4+4

"Book the flight through Houston."

S → NP VP	Det → that this a the
S → Aux NP VP	Noun → book man flight
S → VP	Verb → book include prefer man
NP → Pronoun	
NP → Proper-Noun	Pronoun → I she me
NP → Det Nominal	Proper-Noun → Houston TWA
Nominal → Noun	Aux → does
Nominal → Nominal Noun	Preposition → from to through

Please Turn Over

Nominal → Nominal PP	
VP → Verb	
VP → Verb NP	
VP → Verb NP PP	
VP → Verb PP	
VP → VP PP	
PP → Preposition NP	

8. Consider a corpus consisting of a large set of English sentences. You are tasked with building a trigram language model from this corpus. Answer the following :
- Describe the steps you would take to pre-process the text data before building the trigram model. Include considerations for handling punctuation, casing, and tokenization.
 - Explain how you would construct the trigram language model. Detail the mathematical formulation and how you would estimate the probabilities of trigrams from the given corpus.
 - Discuss at least three different smoothing techniques you could use to handle the issue of zero probabilities for unseen trigrams. Compare their advantages and disadvantages.
 - Suppose you want to improve your trigram model by incorporating additional context, such as part-of-speech tags or named entity recognition. Describe how you would modify your model to include this information and discuss the potential impact on the model's performance. 2+2+3+3
-